



DEALER BEST PRACTICE SERIES

MARC Rate Development

Application Maintenance Site Management Component Rebuild Safety **MARC Management**

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1.0 Introduction

Maintenance and Repair Contracts (MARC) can be both a source of revenue for a dealership and at the same time provide a high level of customer satisfaction. For both of these goals to be reached, it's critical that a dealership uses a cross-functional approach that ensures the input and participation of multiple departments.

Areas such as parts, service, sales, and administration need to be involved in developing the MARC rate. Each area can have critical assumptions that should be involved in rate development.

The *Dealer MARC Rate Development Best Practice* ensures an enterprise communication of roles, responsibilities, and expectations. With these established, the goals of revenue generation and customer satisfaction are easier to reach.

2.0 Best Practice Description

2.1 Evaluate and Understand Customer Needs

A dealer must thoroughly understand the customer's requirements before establishing the per-hour MARC rate. Those requirements must be documented and communicated and become the **"inclusions"** and **"exclusions"** in the per-hour rate.

Among other things, the dealership gathers market intelligence about customer needs, industry trends, political and environmental conditions, and develops an understanding of competitive offerings in the service and parts after-market business.

2.1.1 Inputs

- Dealer/Caterpillar definition of product and market
- Product and market strategies
- Customer contact and interviews
- External research/competition
- Dealer solution history
- Customer fingerprint on current business dealings (i.e. Parts / labor sales)
- Contracts performance
- Special pricing arrangements
- Environmental and political conditions
- Equipment sales
- Economic forecast

2.1.2 Steps

1. View business from customers' perspective
2. Develop list of qualifying and determining needs based on customers' behaviors
3. Force rank all needs in order of importance
4. Capture field data for competitor market penetration and performance
5. Determine competitor capabilities, both current and future
6. Review solution history

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2.1.3 Outputs

- Customer and market needs
- Dealer/competitor strengths and weaknesses (as viewed by market and customer)
- Sales opportunities

2.2 Complete Dealer Capability Review

MARC rate development begins with a required dealer capability review. It compares current dealership capabilities to the customer request for service in the MARC, and documents the gaps. It is important to review and understand exactly what the customer wants and to determine if the dealer can deliver the request. This process needs to be completed up front in order to ensure sufficient time to explore a cost-benefit analysis of the opportunities presented by the customer, or to secure a supplier commitment for the service.

The Dealer Capability Review is broken down into 4 sections:

1. General Contract Information
2. Dealer Workshop Capability
3. Dealer Component Repair and Salvage Capability
4. Dealer Labour Skills Capability

(See attached Dealer Capability Review Form.doc)

2.3 Complete Due Diligence Process

The due diligence process will be the safe source document to capture contract details, critical assumptions, site profile and capabilities, application information, Maintenance and Repair Process responsibilities and component risks. This work is done up front to ensure consistent contract communication across the dealer departments.

For assistance with the Due Diligence process, please contact the CGM Regional MARC Consultant for your region.

(See attached RP2009 Due Diligence Form – revised June 2009.xlsx)

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2.4 Complete CliPP Analysis

The CliPP data sheet is used to record information for the lower power train life prediction. Once the CliPP data sheet has been filled out, please contact the CGM Regional MARC Consultant for your region to have the analysis performed and the results communicated.

(See attached CLIPP Analysis Form.doc)

2.5 Localize the Builder File

Assembling the MARC Team is an important part of “Localizing” the Builder file. It is an important step to ensure the proposed site operations group can execute the MARC for the proposed parts and labor dollars for each repair option: R&I, Exchange Components, Running Repairs, and Preventive Maintenance (PM's).

The perspective must be from whatever facility will be used to deliver the service. Quoted PM times must be in line with the dealer's capability at the Site. For example, if the quoted PM time is 25 hours for a 500-hour truck PM, can it be completed in 25 hours? Site-specific details (does the work need to be done outside, etc.) must be considered. If operations cannot execute a service in the hours listed in Builder, then adjustments **MUST** be made prior to tendering the rate per hour to the customer.

For each Repair Option, compare the proposed intervals with the due diligence document and the territory experience for each model. Make adjustments as required to ensure the total number of repair events is calculated for the requested term.

Search Criteria

Business Group: Family: 22 DHT & TRACTORS/AT

Manufacturer: AA Caterpillar Job:

Model: 793C_HD Component:

S/N Range: ATY 00202 - 99999

Repair Option:

Job Operation:

Repair Options

☒ Customer List
☐ Dealer Net
☐ Gross Profit

◀ ◻ ▶

Parts	Arrangement No	Group No	Comp	Job	Modif	Qc	Description	Lc	App	Cc	First Interval	Next Interval	Quantity	Additions
X	1470797	1470797	1000	007			1470797: ENGINE RECONDITION AFTER FAILURE				16000	16000	1	10R1157
X	1905992	1470797	1000	010			1470797: ENGINE REMOVE & INSTALL				16000	16000	1	
X	1470797	1470797	1000	019			1470797: ENGINE RECON BEFORE FAILURE FIRST OVERHAUL	10HL			16000	100000	1	
X	1470797	1470797	1000	019			1470797: ENGINE RECON BEFORE FAILURE SECOND OVERHAUL	20HL			32000	100000	1	
X	1470797	1470797	1000	513			1470797: ENGINE REPLACE WITH CAT REMAN				48000	16000	1	10R1157
X	1893793	1470797	1000	900			1470797: ENGINE MISCELLANEOUS REPAIRS				500	500	1	
X	1779040	1942975	1052	007	LF		1942975: TURBOCHARGER RECONDITION AFTER FAILURE LEFT FRONT				8000	16000	1	
X	1779040	1942975	1052	007	LR		1942975: TURBOCHARGER RECONDITION AFTER FAILURE LEFT REAR				8000	16000	1	
X	1779040	1942975	1052	007	RF		1942975: TURBOCHARGER RECONDITION AFTER FAILURE RIGHT FRONT				8000	16000	1	
X	1779040	1942975	1052	007	RR		1942975: TURBOCHARGER RECONDITION AFTER FAILURE RIGHT REAR				8000	16000	1	
X	1779040	1942975	1052	010	LF		1942975: TURBOCHARGER REMOVE & INSTALL LEFT FRONT				8000	16000	1	
X	1779040	1942975	1052	010	LR		1942975: TURBOCHARGER REMOVE & INSTALL LEFT REAR				8000	16000	1	
X	1779040	1942975	1052	010	RF		1942975: TURBOCHARGER REMOVE & INSTALL RIGHT FRONT				8000	16000	1	
X	1779040	1942975	1052	010	RR		1942975: TURBOCHARGER REMOVE & INSTALL RIGHT REAR				8000	16000	1	
X	1779040	1942975	1052	019	LF		1942975: TURBOCHARGER RECON BEFORE FAILURE LEFT FRONT				8000	16000	1	
X	1779040	1942975	1052	019	LR		1942975: TURBOCHARGER RECON BEFORE FAILURE LEFT REAR				8000	16000	1	
X	1779040	1942975	1052	019	RF		1942975: TURBOCHARGER RECON BEFORE FAILURE RIGHT FRONT				8000	16000	1	
X	1779040	1942975	1052	019	RR		1942975: TURBOCHARGER RECON BEFORE FAILURE RIGHT REAR				8000	16000	1	
X	1779040	1942975	1052	513	LF		1942975: TURBOCHARGER REPLACE WITH CAT REMAN LEFT FRONT				8000	16000	1	
X	1779040	1942975	1052	513	LR		1942975: TURBOCHARGER REPLACE WITH CAT REMAN LEFT REAR				8000	16000	1	

Total Of Shown Repair Options

Downtime	Labor Hours	Labor	Parts	Miscellaneous	Total
0	400	\$28,000.00	\$116,919.51	\$0.00	\$144,919.51

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Next, for each Repair Option, compare the proposed Parts and Labor dollars to the dealership's ability to execute and deliver the MARC. The objective is to confirm that the operations Project Manager's group can deliver the proposal, based on the customer requirements, dealer capability, and current list of standard jobs. If unfavorable, change the Builder file to match the capability.

(See the Best Practice "Localizing a Centerline BUILDER File for MARC Rate Development" for more detailed instructions.)

2.6 Review Infrastructure Cost to Deliver Required Services

Once repair options have been priced in accordance to the pricing strategy, additional infrastructure costs must be calculated. This includes such factors as:

- Additional tools, equipment, facilities
- Employees or skills required
- Parts capabilities
- Location
- Freight
- Storage
- Contract administration
- Service technicians and classifications
- Management staff
- Logistics (travel time, mileage, housing, meals)
- Communication equipment
- Computer systems.

The infrastructure review may also reveal the customer requires a service on-site on a product currently not supported by the dealership. This process may require additional training to current staff or it may require that another resource with the specific skills be hired by the dealership. These additional infrastructure costs must be included in the final MARC contract. The attached worksheet template can be used to calculate the additional infrastructure costs required to execute the MARC.

(See attached Infrastructure Requirements Form.xls)

2.7 Identify Logistics Costs to Deliver the Required Services

It is essential that any cost related to transporting tools, equipment or staff be accounted for in the final MARC cost model. The dealership can determine if a favorable cost-benefit business opportunity exists. The dealer must determine how to recoup all or part of the investment needed to meet those logistic requirements.

(See attached Logistics Cost Worksheet.xls)

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2.8 Identify Financial Costs to Deliver the Required Services

Each MARC contract will be unique and will require an analysis of additional financial costs and risks the dealership must realize to deliver the proposed services. Financial costs include depreciation and additional insurance expenses, additional tax implications, the cost of money and other liabilities the dealership must identify prior to committing an obligation to deliver a MARC. A worksheet template can be used to calculate the additional financial costs required to execute the MARC.

(See attached Financial Costs Form.doc)

2.9 Document Risk Decisions – Probability & Reserve

Contract risk is documented as part of the due diligence process; however it is very important to calculate the “**cost**” of risk. Risk is defined in several areas of the contract;

2.9.1 Before and After Failure Probability (80/20)

The probability review entails accounting for repair event risk. Some components will “cost” slightly more than the repair option in the business plan, and some will “cost” slightly less. The 80/20 probability process takes 80% of the before-failure repair option price and 20% of the after-failure repair option price, and combines both figures to establish the repair option price.

A “what if” analysis can be performed using “80/20”, “70/30”, or “50/50”. The level of confidence the dealer has in the components will dictate the probability percentage used.

The following example illustrates an engine repair using 80/20 for the first, second and third overhaul, and the remove and install labor to establish a rate per hour for the engine repair option.

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2.9.2 Repair Reserve Contingency

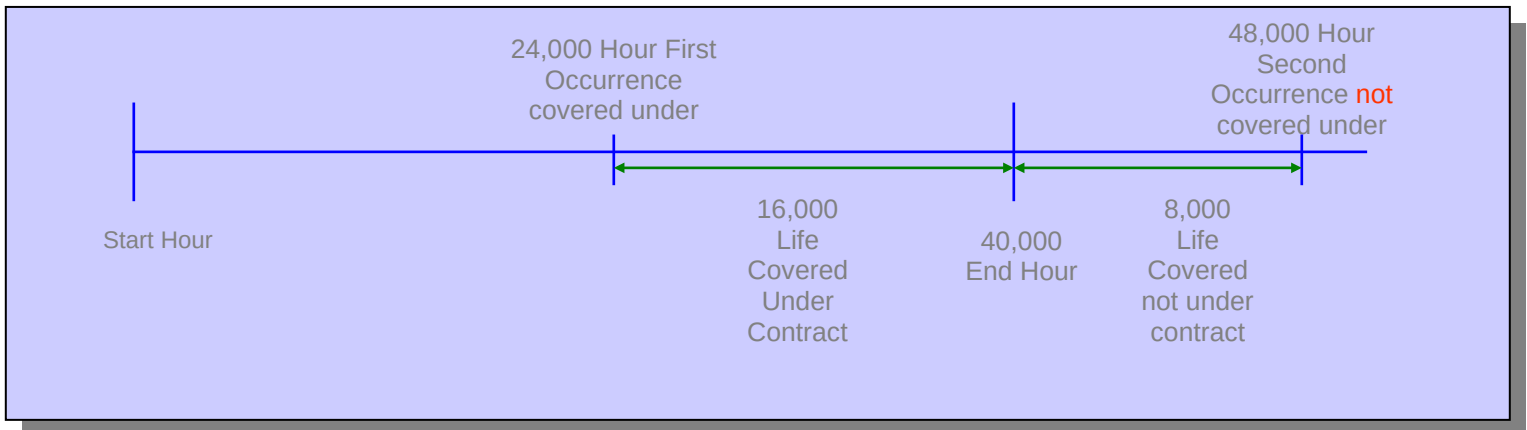
- To build risk into a contract.
- To provide a means for having consistent pricing between an initial MARC sale and subsequent renewals of a contract.

A simple example of “Contingency” is as follows:

- Dealer prices a MARC contract on an engine that covers only a major overhaul. The major overhaul is scheduled to occur every 24,000 hours at a cost of \$100,000. The contract term is 40,000 hours.

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Below is a timeline graph of this scenario:



“Contingency” is defined as the cost of a component that is associated with the percentage of life used while under contract. “Contingency” is only calculated for components that are to be installed during the contract term, but will be removed after the contract has expired.

In the above scenario, the:

- % of Life Used Under the Contract = $16,000/24,000 = 66\%$
- % of Life Used not under the Contract = $8,000/24,000 = 33\%$

In the above scenario, the “Contingency” = 66% of the cost of the component (\$100,000) = \$66,666

Therefore, **“Contingency” = (% of Life Used Under the Contract) * (Cost of Event)**

If “Contingency” were applied to this contract, the cost of the contract would be the sum of the first occurrence of the event under contract at 24,000 hours plus the “Contingency” amount of the second event.

First Event	\$100,000
“Contingency”	<u>\$ 66,666</u>
Total Cost	\$166,666

This scenario can vary depending on the frequency of the event and the term of the contract.

Trigger (%):

In the prior example, the event had a used life of 66% at the end of the contract. When the user enters a percent sign (%) in this field in *CALCULATOR*, the application will only calculate “Contingency” if the used life is greater than the Trigger percentage.

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If, in the above example, the Trigger were set to 70% for this event, "Contingency" would not be calculated and the contract would be priced at \$100,000. If the Trigger were set to 60% for this event, "Contingency" would be calculated and the contract would be priced at \$166,666.

Reserve (%)

Since the Trigger causes an "all or nothing" scenario, an additional adjustment can be made to the "Contingency" formula using a Reserve percentage.

Refer to the original example:

Scenario 1: Trigger = 60% and Used Life = 66% - "Contingency" is calculated as \$66,666

Scenario 2: Trigger = 70% and Used Life = 66% - "Contingency" is calculated as \$0

Suppose in Scenario 2; there were a preference to calculate some "Contingency" (instead of \$0).

If the "Reserve %" were set to 50%, the above example would calculate Contingency as follows:

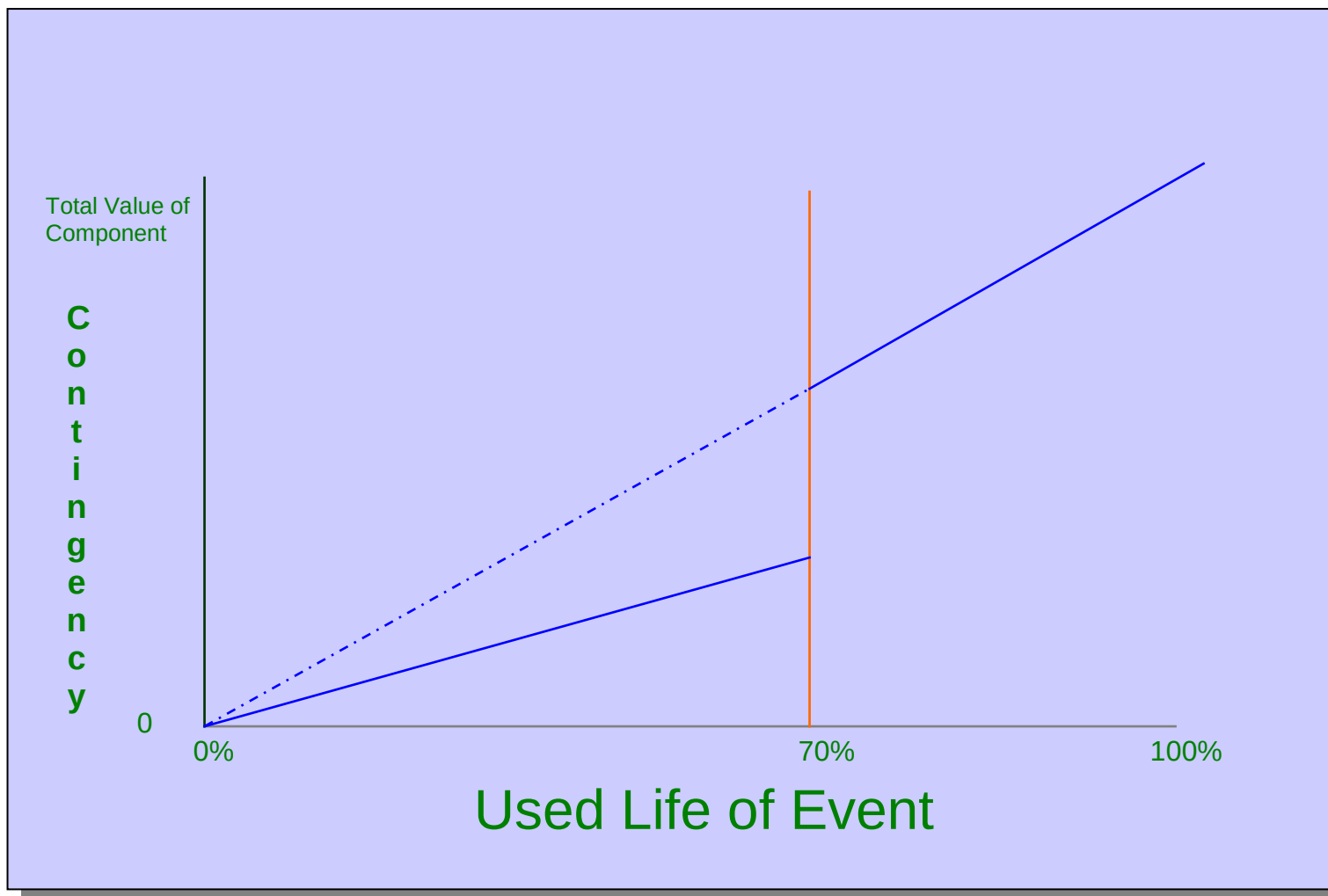
1. Calculate the cost of Contingency,
2. Apply the Trigger of 70%,
3. The Used Life of the event (66%) is below the trigger %,
4. In this case, the Reserve is calculated as 50% of the total "Contingency" or \$33,333
5. Total cost for the contract = \$133,333.

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Contingency Graph

In the graph below, values are shown that could be calculated based on a specific Trigger and Reserve %. Assume that the Trigger is 70% and the Reserve is 50%.



Contingency as Risk

Assume the value (\$100,000) of the event in the example is the cost of a “before failure” repair. This means that the contract was calculated assuming all occurrences of this event are scheduled, before failure events. There is an inherent risk that the event may occur as a failure.

Applying “Contingency” can provide for the risk of failure events. In our example a value of \$66,666 is applied for the risk of failure. This may be a significant cost for a customer, so the user should consider using the Trigger and Reserve % to adjust the risk.

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If the Dealer creates a MARC using “Before Failure” and “After Failure” to provide for risk, the user should be careful when applying “Contingency” to avoid jeopardizing the competitiveness of the rate.

Contingency for Smoothing Contract Extensions

- In the example, the dealer calculates a 40,000-hour contract for \$100,000. The cost-per-hour (CPH) is \$2.50
- If an extension is offered after the end of the initial contract for 8,000 hours. The CPH would be \$12.50
- The reason is: the customer has a shorter time frame (8,000 vs. 40,000 hours) to pay for the Engine.
- If contingency is applied to the first contract. The total is \$166,666, and the CPH would be \$4.17
- On the extension the customer prepays for \$66,666 of the Engine in the first contract. The total of the extension is \$33,333. The CPH is \$4.17

2.9.3 Component Intervals

Consider the number of rebuilds over the MARC term. Selecting appropriate component intervals is an important element of the rate development process. When selecting appropriate component intervals, consider the number of components required as well as how realistic and achievable the interval is.

Small incremental improvements in component life can have a significant impact on the total contract cost. Average Component Lives can impact the total number of components over the contract term.

C O M P O N E N T H O U R S	6000		1	2	3	4	5	6	7	8	9
	8000		1	2	3	4	5	6	7		
	10 000		1	2	3	4	5				
	12 000		1	2	3	4					
	15 000		1	2	3						
	20 000		1	2							
	30 000		1								
		0	10k	20k	30k	40k	50k	60k			
Contract Terms											

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2.10 Develop Rate Per Hour using *Calculator*

Once the due diligence is complete and the infrastructure requirements have been reviewed, *Calculator* can be used to combine repair options and infrastructure in establishing the “rate-per-hour,” to present to the customer.

Calculator contains a complete listing of Repair Options, PM's, Infrastructure, Travel Time & Mileage, and Management Fees that equate to a rate-per-hour.

Additional Note: If Dealer Net pricing is established in Builder for both Parts and Labor, then Calculator can also provide the corresponding Cost per Hour in addition to the MARC Rate per Hour.

Summary		Application		Pricing Adjustments		Advanced Options		General Notes	
Estimate Name	Arch Coal	Unit of Measure		Start	0				
Version Description	BUILDER Import	Usage - Hour	1.0000	End	60000				
Linked to a Contract		Serial No Range	FDB00001 - 99999						
Export to Sales Tool ☐					Finalized ☐				
					View Sell				
Inc	Grouping	Sell	Cont Sell	Total Sell	Gross Margin	Rate Sell	USD/	Labor	Downtime
☒	Engine	\$966,172	\$0	\$966,172	24.31%	\$16.10	\$16.10	2,728.7	1,610.0
☒	Powertrain	\$1,048,054	\$0	\$1,048,054	17.17%	\$17.47	\$17.47	1,719.4	600.0
☒	Other Components	\$0	\$0	\$0	0.00%	\$0.00	\$0.00	0.0	0.0
☒	Hydraulics	\$352,108	\$0	\$352,108	27.83%	\$5.87	\$5.87	949.9	731.0
☒	GET/Undercarriage	\$16,215	\$0	\$16,215	-5.35%	\$0.27	\$0.27	136.0	48.0
☒	Attachments	\$211,513	\$0	\$211,513	-2.29%	\$3.53	\$3.53	1,321.0	799.0
☒	Maintenance	\$1,108,279	\$0	\$1,108,279	34.39%	\$18.47	\$18.47	4,562.0	2,696.0
☒	General	\$0	\$0	\$0	0.00%	\$0.00	\$0.00	0.0	0.0
☒	Travel	\$465,677	\$0	\$465,677	56.40%	\$7.76	\$7.76	4,414.0	0.0
☒	Mgmt & Misc	\$0	\$0	\$0	0.00%	\$0.00	\$0.00	0.0	0.0
☐	R&M Total	\$4,168,019	\$0	\$4,168,019	27.61%	\$69.47	\$69.47	15,830.9	6,484.0
☒	Other Ownership Costs	\$0	\$0	\$0	0.00%	\$0.00	\$0.00	0.0	0.0
☒	Estimate Total	\$4,168,019	\$0	\$4,168,019	27.61%	\$69.47	\$69.47	15,830.9	6,484.0

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3.0 Implementation Steps

Each of the steps explained above.

1. Evaluate and understand customer needs
2. Dealer Capability Review
3. Complete due diligence process
4. *ClIPP* Analysis
5. CGM Builder Centerline file (refer to *CGM Builder Best Practice*)
6. Localize Builder file
7. Review Infrastructure costs to deliver required services
8. Identify logistics costs to deliver required services
9. Identify financial costs to deliver required services
10. Document Risk Decisions – Probability & Reserve
11. Create Calculator Budget

4.0 Benefits

- Predictable revenue stream to dealer
- Enterprise approach to the Mining MARC opportunity.
- Employee satisfaction will result from participating in the rate development process and clearly communicated goals and expectations.
- Customer satisfaction resulting from proper dealer execution and communication. Ensuring MARC team members understand their roles and expectations, execution is a simply a function of extending authority and allowing the staff to be responsible for their own areas.

5.0 Resources Required

- The primary resource required to implement the Rate Development Process is an enterprise approach to MARC management. It is imperative to approach the MARC management business from top management down. Top-down directives ensure the staff is empowered to make business decisions and has the authority and management support required to be successful.
- Each dealer department will need a dedicated resource as a representative to support the project sponsor and MARC manager when implementing an enterprise Rate Development Process.
- *Builder* and *Calculator* software.

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6.0 Supporting Attachments / References

- [CLIPP Analysis Form.doc](#) (MS-Word document)
- [Infrastructure Requirements Form.xls](#) (MS-Excel document)
- [Logistics Cost Worksheet.xls](#) (MS-Excel document)
- [Financial Cost Form.doc](#) (MS-Word document)
- [Dealer Capability Review Form.doc](#) (MS-Word Document)
- [RP2009 Due Diligence Form – Revised June 2009.xlsx](#) (MS-Excel document)

7.0 Related Best Practices

0906-3.3-1021 - Developing a MARC Management Plan

0907-3.3-1086 - Localizing a Centerline BUILDER File for MARC Rate Development

8.0 Acknowledgements

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